

MATH 2413 Differential Calculus

Lecture 1: Pre-Calculus Review-I

Factoring Formulas

(a) $(a + b)^2 = a^2 + 2ab + b^2$ and $(a - b)^2 = a^2 - 2ab + b^2$

(b) $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$ and $(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$

(c) $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$ and $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$

(d) $a^2 - b^2 = (a - b)(a + b)$

Example 1 *Factor Completely*

(a) $x^3 - 27 =$

(b) $x^3 + 125 =$

Zero Product Rule

For any real numbers x and y :

$$xy = 0 \text{ if and only if } x = 0 \text{ or } y = 0.$$

Quadratic Equation

The solutions to $ax^2 + bx + c = 0$, where $a \neq 0$ are given by

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

Example 2 *Solve the following equations:*

(a) $2x^2 - 5x + 2 = 0$

(b) $(2x + 1)(x - 3) = -6$

(c) $(2x + 1)(x - 3) = 0$

Absolute Value

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

Example 3 Simplify $\frac{|x-2|}{x-2}$ if

(a) $x = 2$

(b) $x > 2$

(c) $x < 2$

Exponential Function

A function of the form $f(x) = a^x$ ($a > 0, a \neq 1$) is called an **exponential function** with base a .

Example 4 Graph the exponential functions $f(x) = 2^x$ and $g(x) = \left(\frac{1}{2}\right)^x$. State the domain and range.

Natural Exponential Function

The exponential function e^x with base $e \approx 2.7183$ is called the *natural exponential function*.

Logarithmic Functions

For $a > 0$, $a \neq 1$ and $y > 0$ define

$$x = \log_a(y) \quad \text{iff} \quad a^x = y$$

and we call the real number x the logarithm of y with base a . By definition, it is clear that an exponential function with base a and a logarithmic function with a are inverse of each other. If $a = e \approx 2.7183$, then

$$\ln(y) = \log_e(y)$$

is called the **natural logarithm** of y .

Example 5 Graph $y = \log_2 x$ and $y = \log_{\frac{1}{2}} x$. What are the domain and the range?

Properties of Logarithms

Let $a > 0$ and $a \neq 1$, then for all positive x, y ,

1. $\log_a(xy) = \log_a(x) + \log_a(y)$
2. $\log_a\left(\frac{x}{y}\right) = \log_a(x) - \log_a(y)$
3. $\log_a(x^r) = r \log_a(x)$, r is any real number
4. $\log_a x = \frac{\ln x}{\ln a}$

Example 6 Use the properties of logarithms to expand the expression completely

$$\ln \left[\frac{(x^2 + 3)^4}{x \sqrt[3]{x^2 + 1}} \right].$$

Example 7 *Graph the functions. Write the domain and range*

(a) $\log_2(x + 2)$

(b) $\log_2(x - 2)$

(c) $\log_2(x) + 2$

(d) $\log_2(x) - 2$

Lecture 2: PreCalculus Review-II

Trigonometric Identities

(a) $\sin^2(\theta) + \cos^2(\theta) = 1$

(b) $1 + \tan^2 \theta = \sec^2 \theta$

(c) $1 + \cot^2 \theta = \csc^2 \theta$

(d) $\sin(\alpha \pm \beta) = \sin(\alpha) \cos(\beta) \pm \sin(\beta) \cos(\alpha)$

(e) $\cos(\alpha \pm \beta) = \cos(\alpha) \cos(\beta) \mp \sin(\alpha) \sin(\beta)$

(f) $\tan(\alpha \pm \beta) = \frac{\tan(\alpha) \pm \tan(\beta)}{1 \mp \tan(\alpha) \tan(\beta)}$

(g) $\sin(2\theta) = 2 \sin(\theta) \cos(\theta)$

(h) $\sin^2(\theta) = \frac{1}{2}(1 - \cos(2\theta))$

(i) $\cos(2\theta) = \cos^2(\theta) - \sin^2(\theta)$

(j) $\cos^2(\theta) = \frac{1}{2}(1 + \cos(2\theta))$

(k) $\tan(2\theta) = \frac{2 \tan(\theta)}{1 - \tan^2(\theta)}$

(l) $\sin \alpha + \sin \beta = 2 \sin\left(\frac{\alpha + \beta}{2}\right) \cos\left(\frac{\alpha - \beta}{2}\right)$

(m) $\sin \alpha - \sin \beta = 2 \sin\left(\frac{\alpha - \beta}{2}\right) \cos\left(\frac{\alpha + \beta}{2}\right)$

(n) $\cos \alpha + \cos \beta = 2 \cos\left(\frac{\alpha + \beta}{2}\right) \cos\left(\frac{\alpha - \beta}{2}\right)$

(o) $\cos \alpha - \cos \beta = -2 \sin\left(\frac{\alpha + \beta}{2}\right) \sin\left(\frac{\alpha - \beta}{2}\right)$

(p) $\sin \alpha \sin \beta = \frac{1}{2}(\sin(\alpha + \beta) - \sin(\alpha - \beta))$

(q) $\cos \alpha \cos \beta = \frac{1}{2}(\cos(\alpha + \beta) + \cos(\alpha - \beta))$

(r) $\sin \alpha \cos \beta = \frac{1}{2}(\sin(\alpha + \beta) + \sin(\alpha - \beta))$

Example 8 *Solve the following trigonometric equations*

(a) $3 \tan^2 2x - 1 = 0$

(b) $2 \sin^2 x - \sin x - 1 = 0$

(c) $2 \sin x + \csc x = 0$

(d) $\sec x + \tan x = 1$

Inverse Trigonometric Functions

Example 9 Graph the functions $f(x) = \sin x$ and $g(x) = \sin x$ with domain $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

From the graph, you must have realized that the function $f(x) = \sin x$ is not one to one but the function $g(x) = \sin x$ with domain $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ is one to one and hence has inverse. This inverse is called the *inverse sine* function and is written as $\sin^{-1} x$ or $\arcsin x$. Therefore

$$y = \sin^{-1} x \iff x = \sin y, \quad -\frac{\pi}{2} \leq y \leq \frac{\pi}{2}, \quad -1 \leq x \leq 1.$$

Example 10 Graph the function $\arcsin x$

Example 11 *Repeat the above for other 5 inverse trigonometric functions.*

Example 12 Find the following values of compositions

(a) $\sin^{-1} \left(\sin \left(\frac{5\pi}{4} \right) \right)$

(b) $\cos^{-1} \left(\cos \left(\frac{7\pi}{6} \right) \right)$

(c) $\cos^{-1} \left(\sin \left(\frac{7\pi}{6} \right) \right)$

(d) $\sin \left(\cos^{-1} \left(\frac{\sqrt{2}}{2} \right) \right)$

(e) $\cos \left(\sin^{-1} \left(-\frac{\sqrt{3}}{2} \right) \right)$

(f) $\tan \left(\cos^{-1} \left(\frac{1}{2} \right) \right) =$

$$(g) \sin \left(\sin^{-1} \left(\frac{\sqrt{2}}{2} \right) + \cos^{-1} (1) \right)$$

$$(h) \cos \left(\sin^{-1} \left(-\frac{3}{5} \right) - \cos^{-1} \left(\frac{4}{5} \right) \right)$$

$$(i) \sin \left(2 \cos^{-1} \left(\frac{4}{5} \right) \right)$$

Example 13 Let $\cos \theta = \frac{4}{5}$, where $\frac{3\pi}{2} < \theta < 2\pi$. Find $\sin(2\theta)$, $\cos(2\theta)$, $\sin\left(\frac{\theta}{2}\right)$, $\cos\left(\frac{\theta}{2}\right)$.

Example 14 If $f(x) = \arctan x$, compute the number

$$\frac{f(\sqrt{3}) - f(1)}{\sqrt{3} - 1}$$

round your result to two decimal places.

Lecture 3: Limit of a Function

Notation:

- $x \rightarrow 2^+$ means x approaches to 2 from the right.
- $x \rightarrow 2^-$ means x approaches to 2 from the left.
- $x \rightarrow 2$ means x approaches to 2 from either side.

Example 15 Let $f(x) = \frac{3x^2 - 6x}{x - 2}$, complete the following table

x	1.9	1.99	1.999	
$f(x)$				

Estimate the following limits

(a) $\lim_{x \rightarrow 2^-} f(x) =$

(b) $\lim_{x \rightarrow 2^+} f(x) =$

(c) $\lim_{x \rightarrow 2} f(x) =$

Informal Definition of Limit:

Suppose that f is defined in a *deleted neighborhood* of 'a' (an open interval containing 'a' excluding 'a' itself), then $f(x)$ is said to have the limit L at $x = a$ (or as x approaches a), and we write

$$\lim_{x \rightarrow a} f(x) = L$$

if the values $f(x)$ can be made arbitrarily close to L by taking x sufficiently close to a but not equal to a

Example 16 Let $f(x) = \frac{x^3 - 1}{x - 1}$, complete the following table and estimate $\lim_{x \rightarrow 1} f(x)$. Is $f(x)$ defined at 1?

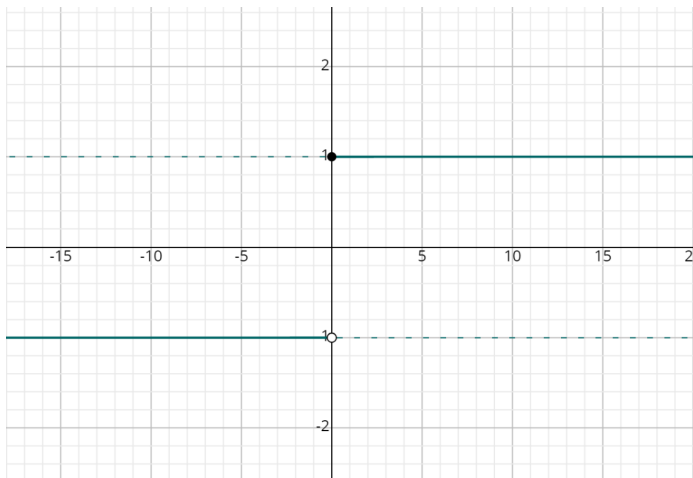
x	0.9	0.99	0.999	1	1.0001	1.001	1.01	1.1
$f(x)$								

Note: The method we used to estimate the limit in previous 2 examples is called the *numerical method*.

Limits that fail to exist:

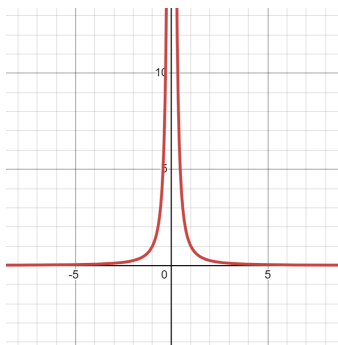
1. Different Right and Left Behavior: Consider the function

$$f(x) = \begin{cases} -1 & \text{if } x < 0 \\ 1 & \text{if } x \geq 0 \end{cases}$$



Then $\lim_{x \rightarrow 0^-} f(x) = -1$ but $\lim_{x \rightarrow 0^+} f(x) = 1$. Here, $\lim_{x \rightarrow 0^-} f(x) \neq \lim_{x \rightarrow 0^+} f(x)$, therefore $\lim_{x \rightarrow 0} f(x)$ does not exist.

2. Unbounded Behavior: If $f(x)$ keeps growing in magnitude without a bound as $x \rightarrow a$, then $\lim_{x \rightarrow a} f(x)$ does not exist.

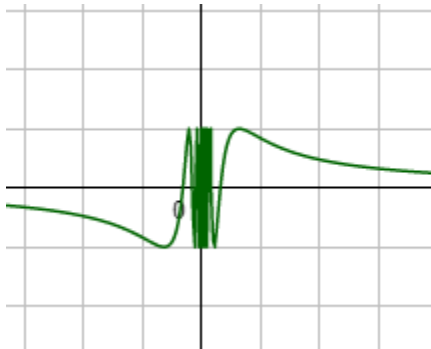


$$\lim_{x \rightarrow 0} \frac{1}{x^2} = DNE.$$

3. Oscillation: Consider the function $f(x) = \sin\left(\frac{1}{x}\right)$, then

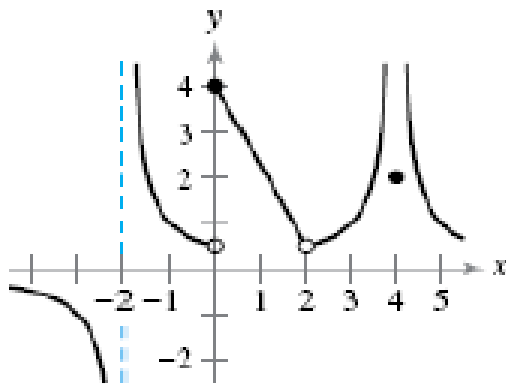
x	$\frac{2}{\pi}$	$\frac{2}{3\pi}$	$\frac{2}{5\pi}$	$\frac{2}{7\pi}$	$\frac{2}{9\pi}$	$\frac{2}{11\pi}$	$\frac{2}{13\pi}$	$\frac{2}{15\pi}$
f(x)	1	-1	1	-1	1	-1	1	-1

As we get closer and closer to 0 the function does not approach to any number but keeps oscillating between 1 and -1 . The graph looks like below



Therefore, $\lim_{x \rightarrow 0} \sin\left(\frac{1}{x}\right)$ does not exist.

Example 17 Graph of $y = f(x)$ given, evaluate the following.



- | | | | |
|---------------------------------------|-----------------------------------|---------------------------------|----------|
| (a) $\lim_{x \rightarrow 2^-} f(x) =$ | $\lim_{x \rightarrow 2^+} f(x) =$ | $\lim_{x \rightarrow 2} f(x) =$ | $f(2) =$ |
| (b) $\lim_{x \rightarrow 0^-} f(x) =$ | $\lim_{x \rightarrow 0^+} f(x) =$ | $\lim_{x \rightarrow 0} f(x) =$ | $f(0) =$ |
| (c) $\lim_{x \rightarrow 1^-} f(x) =$ | $\lim_{x \rightarrow 1^+} f(x) =$ | $\lim_{x \rightarrow 1} f(x) =$ | $f(1) =$ |
| (d) $\lim_{x \rightarrow -2} f(x) =$ | $\lim_{x \rightarrow 4} f(x) =$ | $f(4) =$ | |

Note: In example 3, we found limits graphically.

Evaluating Limits Analytically

Let $f(x)$ and $g(x)$ be defined for all x in a deleted neighborhood I of c and $f(x) = g(x)$ in I , then

$$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} g(x).$$

Example 18 *Find the following limits*

(a) $\lim_{x \rightarrow 3} \frac{x^3 - 27}{x^2 - 5x + 6}$

(b) $\lim_{x \rightarrow 3} \frac{\sqrt{3x + 7} - 4}{x - 3}$

Lecture 4: Calculating Limits

Theorem 1 (Some Basic Limits) For any real numbers a, b ;

$$(a) \lim_{x \rightarrow a} b = b$$

$$(b) \lim_{x \rightarrow a} x = a$$

Example 19

$$(a) \lim_{x \rightarrow 99} 5 =$$

$$(b) \lim_{x \rightarrow 6} x =$$

Theorem 2 (Properties of Limits) Suppose $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$, then

$$(a) \lim_{x \rightarrow a} [cf(x)] = cL \text{ for any constant } c$$

$$(b) \lim_{x \rightarrow a} [f(x) \pm g(x)] = L \pm M$$

$$(c) \lim_{x \rightarrow a} [f(x) \cdot g(x)] = LM$$

$$(d) \lim_{x \rightarrow a} \left[\frac{f(x)}{g(x)} \right] = \frac{L}{M} \text{ provided, } M \neq 0.$$

$$(e) \lim_{x \rightarrow a} [f(x)]^n = L^n \text{ for any positive integer.}$$

$$(f) \lim_{x \rightarrow a} \sqrt[n]{f(x)} = \begin{cases} \sqrt[n]{L}, & \text{if } L \geq 0 \\ \sqrt[n]{L}, & \text{if } n \text{ is odd} \\ DNE & \text{if } n \text{ is even and } L < 0 \end{cases}$$

Note: Theorem 1-2 are also true if $x \rightarrow a$ is replaced by $x \rightarrow a^\pm$.

Example 20 Given $\lim_{x \rightarrow a} f(x) = 4$ and $\lim_{x \rightarrow a} g(x) = -27$, find

$$(a) \lim_{x \rightarrow a} [f(x) - 2g(x)] =$$

$$(b) \lim_{x \rightarrow a} \sqrt{f(x)} =$$

$$(c) \lim_{x \rightarrow a} \sqrt{g(x)} =$$

$$(d) \lim_{x \rightarrow a} \sqrt[3]{g(x)} =$$

$$(e) \lim_{x \rightarrow a} \left[\frac{2f(x) + x + 3}{3g(x) - 2f(x)^2} \right] =$$

$$(f) \lim_{x \rightarrow a} \left[\frac{(f(x))^3}{2g(x) + 4} \right] =$$

Additional Basic Limits

1. $\lim_{x \rightarrow a} x^n = a^n$
2. (Direct Substitution) Let $f(x)$ be a polynomial or rational or trigonometric or logarithmic or exponential or absolute value or inverse trigonometric function and a is in the domain of $f(x)$ then;

$$\lim_{x \rightarrow a} f(x) = f(a)$$

Example 21 Find the following limits:

$$(a) \lim_{x \rightarrow \frac{\pi}{2}} \sin x =$$

$$(b) \lim_{x \rightarrow \frac{\pi}{2}} \cos x =$$

$$(c) \lim_{x \rightarrow \frac{\pi}{2}} \tan x =$$

$$(d) \lim_{x \rightarrow \frac{\pi}{3}} \tan x =$$

$$(e) \lim_{x \rightarrow 1} 2(3x^3 + 5x^2 - 9x + 2)e^{x-1} =$$

$$(f) \lim_{x \rightarrow 1} (x+1)^2 \log_2(x+1) =$$

$$(g) \lim_{x \rightarrow -\frac{1}{2}} [\sin^{-1} x + \cos^{-1} x] =$$

Theorem 3 Let $f(x) = g(x)$ for all x in a deleted neighborhood of a , then $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x)$

Example 22 Find the following limits:

$$(a) \lim_{x \rightarrow 2} \frac{\frac{1}{x} - \frac{1}{2}}{x - 2}$$

$$(b) \lim_{x \rightarrow 2} \frac{x^2 + 3x + 4}{x^2 - 9}$$

$$(c) \lim_{x \rightarrow 2} \frac{x^2 - 5x + 6}{x^2 - 4}$$

$$(d) \lim_{x \rightarrow 2} \frac{x^5 - 32}{x - 2}$$

$$(e) \lim_{x \rightarrow 9} \frac{\sqrt{x} - 3}{x - 9}$$

$$(f) \lim_{x \rightarrow 4} \frac{\sqrt{x + 5} - 3}{x - 4}$$

Example 23 *Let*

$$f(x) = \begin{cases} x^3 - 2x - 1 & \text{if } x \leq -1 \\ x^2 - 1 & \text{if } -1 < x \leq 2 \\ \sqrt{2x} + \sqrt{3x-1} & \text{if } x > 2 \end{cases}$$

find $\lim_{x \rightarrow -1} f(x)$ *and* $\lim_{x \rightarrow 2} f(x)$.

Lecture 5: Infinite Limits

Let $f(x) = \frac{N(x)}{D(x)}$

1. if $\lim_{x \rightarrow a^-} N(x) = L \neq 0$ and $\lim_{x \rightarrow a^-} D(x) = 0$, then

$$\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^-} \frac{N(x)}{D(x)} = \pm\infty.$$

The sign will be determined by the behavior of $N(x)$ and $D(x)$ as $x \rightarrow a^-$.

2. if $\lim_{x \rightarrow a^+} N(x) = L \neq 0$ and $\lim_{x \rightarrow a^+} D(x) = 0$, then

$$\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^+} \frac{N(x)}{D(x)} = \pm\infty.$$

The sign will be determined by the behavior of $N(x)$ and $D(x)$ as $x \rightarrow a^+$.

3. $\lim_{x \rightarrow a} f(x) = \infty$ means $\lim_{x \rightarrow a^-} f(x) = \infty$ and $\lim_{x \rightarrow a^+} f(x) = \infty$
4. $\lim_{x \rightarrow a} f(x) = -\infty$ means $\lim_{x \rightarrow a^-} f(x) = -\infty$ and $\lim_{x \rightarrow a^+} f(x) = -\infty$

Example 24 Evaluate the following limits

(a) $\lim_{x \rightarrow 2^-} \frac{3}{x-2}$

(b) $\lim_{x \rightarrow 2^+} \frac{3}{x-2}$

(c) $\lim_{x \rightarrow 2} \frac{3}{x-2}$

(d) $\lim_{x \rightarrow 2^-} \frac{3}{(x-2)^2}$

(e) $\lim_{x \rightarrow 2^+} \frac{3}{(x-2)^2}$

(f) $\lim_{x \rightarrow 2} \frac{3}{(x-2)^2}$

Vertical Asymptote

A function $f(x)$ (graph of $f(x)$) is said to have a *vertical asymptote* $x = a$

if either $\lim_{x \rightarrow a^-} f(x) = \pm\infty$ or $\lim_{x \rightarrow a^+} f(x) = \pm\infty$.

Example 25 Find all vertical asymptotes to the graph of

$$f(x) = \frac{4x^2 + 4x - 24}{x^4 - 2x^3 - 9x^2 + 18x}.$$

At each vertical asymptote $x = a$, evaluate $\lim_{x \rightarrow a^-} f(x)$, $\lim_{x \rightarrow a^+} f(x)$, and $\lim_{x \rightarrow a} f(x)$.

Example 26 Find all vertical asymptotes to the graph of

(a) $f(x) = \ln(x - 3)$

(b) $f(x) = \tan x$

(c) $f(x) = \sec x$

Lecture 6: Limits of Trigonometric Functions

Theorem 4 (Squeeze Theorem) *Let $h(x) \leq f(x) \leq g(x)$ for all x near 'a' (except possibly at 'a' itself) and*

$$\lim_{x \rightarrow a} h(x) = \lim_{x \rightarrow a} g(x) = L, \text{ then } \lim_{x \rightarrow a} f(x) = L.$$

Example 27 *Evaluate the following limits:*

$$(a) \lim_{x \rightarrow 0} \left[|x| \sin \left(3 + \frac{2}{x} \right) \right]$$

$$(b) \lim_{x \rightarrow 0} \left[x \sin \left(3 + \frac{2}{x} \right) \right]$$

Theorem 5 *When x is measured in radians, then*

$$(a) \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \text{ and}$$

$$(b) \lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0.$$

Corollary 1 *If x is measured in radians and $\lim_{x \rightarrow a} f(x) = 0$, then*

$$(a) \lim_{x \rightarrow a} \frac{\sin[f(x)]}{f(x)} = 1 \text{ and}$$

$$(b) \lim_{x \rightarrow a} \frac{1 - \cos[f(x)]}{f(x)} = 0$$

Example 28 *Evaluate the following limits:*

$$(a) \lim_{x \rightarrow 0} \frac{\sin 5x}{x}$$

$$(b) \lim_{x \rightarrow 0} \frac{\sin 5x}{\sin 7x}$$

$$(c) \lim_{x \rightarrow 0} \frac{\sin^2 5x}{x^2}$$

$$(d) \lim_{x \rightarrow 3} \frac{\sin(x^2 - 9)}{x - 3}$$

$$(e) \lim_{x \rightarrow 0} \frac{\sin 5x}{\tan 7x}$$

$$(f) \lim_{x \rightarrow 0} \frac{2 \sin 5x - \sin 10x}{3x^2}$$

Lecture 7: Precise Definition of a Limit

Precise Definition of Limit

Let f be a function defined on an open interval containing a (except possibly at a itself). Then, $f(x)$ is said to have the limit L as x approaches to a and we write

$$\lim_{x \rightarrow a} f(x) = L$$

if for every $\epsilon > 0$ there is a $\delta > 0$, such that

$$|f(x) - L| < \epsilon \text{ whenever } 0 < |x - a| < \delta$$

Example 29 Use $\varepsilon - \delta$ definition of the limit to prove that $\lim_{x \rightarrow 3} (2x - 5) = 1$.

Example 30 Given $\lim_{x \rightarrow -3} (6 - 5x) = 21$, find a $\delta > 0$ such that $|(6 - 5x) - 21| < 0.023$, whenever $0 < |x - (-3)| < \delta$.

Lecture 8: Continuity

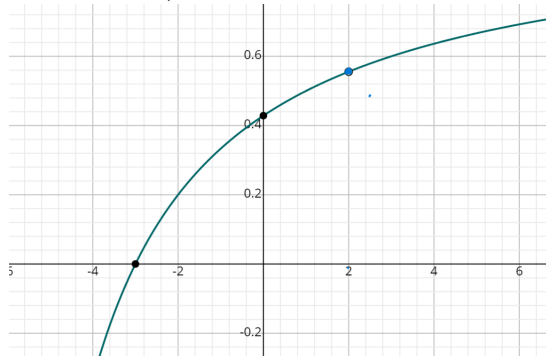
Continuity

A function $f(x)$ is said to be continuous at $x = a$ if

- (i) $f(a)$ is defined
- (ii) $\lim_{x \rightarrow a} f(x)$ exists, and
- (iii) $\lim_{x \rightarrow a} f(x) = f(a)$

If f is not continuous at $x = a$, we say f is discontinuous at $x = a$.

The continuity conditions tell us that if $f(x)$ is continuous at $x = a$, then the graph of $f(x)$ does not break at $x = a$ (can not have gap, jump or hole at $x = a$).



Example 31 Test the continuity of the following functions

(a) $f(x) = 4x^2 + 5x + 6$ at $x = 1$

(b) $f(x) = \sin x$ at $x = 0$

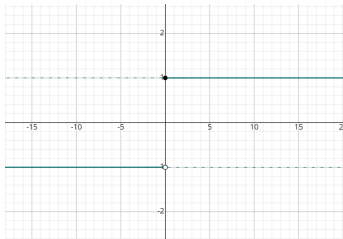
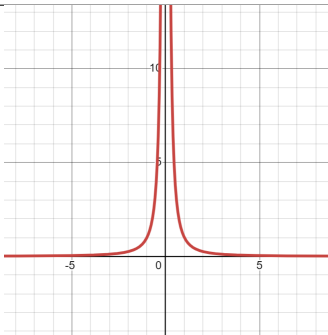
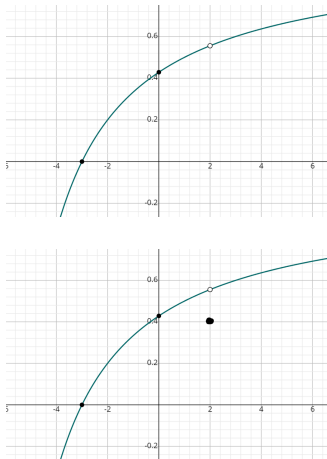
$$(c) \ f(x) = \frac{\sin x}{x} \text{ at } x = 0$$

$$(d) \ f(x) = \frac{x^2 - 5x + 6}{x^2 - 4} \text{ at } x = 2$$

$$(e) \ f(x) = \begin{cases} \frac{\sin x}{x} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases} \text{ at } x = 0$$

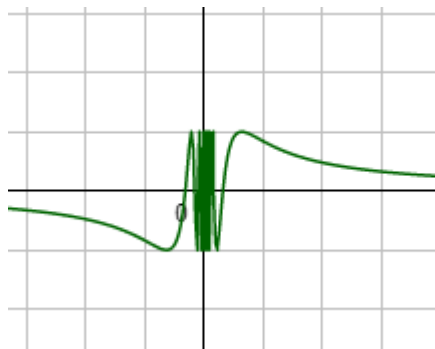
$$(f) \ f(x) = \begin{cases} \frac{5x + \sin 2x}{x} & \text{if } x < 0 \\ 7 & \text{if } x = 0 \\ \frac{x^3 + 7x}{x^4 + 2x} & \text{if } x > 0 \end{cases} \quad \text{at } x = 0$$

Types of Discontinuity

Jump Discontinuity	Infinite Discontinuity	Removable Discontinuity
$\lim_{x \rightarrow a^-} f(x) \neq \lim_{x \rightarrow a^+} f(x)$	$\lim_{x \rightarrow a^-} f(x) = \pm\infty$ or $\lim_{x \rightarrow a^+} f(x) = \pm\infty$	$\lim_{x \rightarrow a} f(x)$ exists but $f(a)$ is undefined or $f(a) \neq \lim_{x \rightarrow a} f(x)$
		

A **removable discontinuity** can be removed by changing or assigning the value of $f(x)$ at $x = a$ to be $\lim_{x \rightarrow a} f(x)$. If a discontinuity at $x = a$ can not be removed by changing or assigning the value of $f(x)$ at

$x = a$, we call it an *essential discontinuity*. Both the jump and infinite discontinuity are essential. The discontinuity of $\sin\left(\frac{1}{x}\right)$ at $x = 0$ is also an essential discontinuity.



Example 32 Test the continuity of $f(x)$ at the given point. If a function is discontinuous, classify the discontinuity. If the continuity is removable, remove it.

(a) $f(x) = \frac{|x-1|}{x-1}$ at $x = 1$

(b) $f(x) = \frac{3}{(x-2)^2}$ at $x = 2$.

(c) $f(x) = \frac{\sin x}{x}$ at $x = 0$

(d) $f(x) = \frac{x^2 - 5x + 6}{x^2 - 4}$ at $x = 2$

$$(e) \ f(x) = \begin{cases} \frac{x^2 - 5x + 6}{x^2 - 4} & \text{if } x \neq 2 \\ 5 & \text{if } x = 2 \end{cases} \quad \text{at } x = 2.$$

$$(f) \ f(x) = \begin{cases} \frac{5x + \sin 2x}{x} & \text{if } x < 0 \\ 5 & \text{if } x = 0 \\ \frac{x^3 + 7x}{x^4 + 2x} & \text{if } x > 0 \end{cases} \quad \text{at } x = 0$$

Intervals of Continuity

1. A function $f(x)$ is continuous on an open interval (a, b) if it is continuous at every point on (a, b) .
2. A function $f(x)$ is continuous on half open interval $[a, b)$ if it is continuous on (a, b) and $\lim_{x \rightarrow a^+} f(x) = f(a)$
3. A function $f(x)$ is continuous on half open interval $(a, b]$ if it is continuous on (a, b) and $\lim_{x \rightarrow b^-} f(x) = f(b)$.
4. A function $f(x)$ is continuous on a closed interval $[a, b]$ if it is continuous on (a, b) , $\lim_{x \rightarrow a^+} f(x) = f(a)$, and $\lim_{x \rightarrow b^-} f(x) = f(b)$.

Lecture 9: Continuity-II

Theorem 6 *If $f(x)$ and $g(x)$ are continuous at $x = a$, then $f(x) \pm g(x)$, $f(x)g(x)$, and $cf(x)$ are continuous at $x = a$. The quotient $\frac{f(x)}{g(x)}$ is also continuous at $x = a$ whenever $g(a) \neq 0$.*

Theorem 7 *Polynomial, rational, trigonometric, inverse trigonometric, exponential, logarithmic, absolute value, and square root functions are continuous on their domains.*

Theorem 8 *If $g(x)$ is continuous at 'a' and $f(x)$ is continuous at ' $g(a)$ ', then ,*

$$(f \circ g)(x) = f(g(x)) \text{ is continuous at } a.$$

Theorem 9 *If $\lim_{x \rightarrow a} g(x) = L$ and $f(x)$ is continuous at L , then*

$$\lim_{x \rightarrow a} f(g(x)) = f(L) = f\left(\lim_{x \rightarrow a} g(x)\right).$$

Example 33 Find the following limits:

$$(a) \lim_{x \rightarrow 1} \tan \left[\frac{\pi}{x+3} + \arcsin(\ln x) \right]$$

$$(b) \lim_{x \rightarrow 3} \arcsin \left(\frac{x^2 - 3x}{x^2 - 9} \right)$$

$$(c) \lim_{x \rightarrow 4} \ln \left(\frac{\sqrt{x} - 2}{x - 4} \right)$$

Example 34 Find the interval of continuity.

$$(a) \ f(x) = \frac{x^2 - 5x + 6}{x^2 - 4}$$

$$(b) \ f(x) = \frac{\sqrt{x-5}}{x^2 - 4}$$

$$(c) \ f(x) = \frac{\sqrt{x-5}}{x^2 - 36}$$

$$(d) \ f(x) = \begin{cases} \frac{\sin x + x}{x} & \text{if } x < 0 \\ 2 & \text{if } x = 0 \\ x^2 + 1 & \text{if } x > 0 \end{cases}$$

$$(e) \ f(x) = e^x + \sqrt[3]{x}$$

$$(f) \ f(x) = \frac{x \ln x}{x - 1}$$

$$(g) \ f(x) = \ln(x^2 + 2)$$

$$(h) \ f(x) = \frac{\arctan x}{e^x - 1}$$

Example 35 Determine the value(s) of 'a' so that

$$f(x) = \begin{cases} 2ax + 4 & \text{if } x < 1 \\ x^2 + ax & \text{if } x \geq 1 \end{cases}$$

is continuous on the entire real line.

Example 36 Find all values of c so that $f(x) = \frac{3x + 5}{x^2 + 6x + c}$ is continuous everywhere.

Theorem 10 (Intermediate Value Theorem (IVT)) *Let*

(a) $f(x)$ be continuous on $[a, b]$

(b) $f(a) \neq f(b)$, and

(c) k lies in between $f(a)$ and $f(b)$,

then there exists a number c in (a, b) such that $f(c) = k$

Example 37 *Verify that the IVT applies. Find a ‘ c ’ guaranteed by the theorem*

$$f(x) = x^3 - x^2 + x - 2, \quad [0, 3], \quad k = 4.$$

Example 38 *Prove that the equation $x^3 + 2x^2 - 4x + 6 = 0$ has a solution on $(-4, 0)$.*

Lecture 10: Limits at Infinity

Limit at Infinity

Let L be a real number. We say that

$$\lim_{x \rightarrow \infty} f(x) = L$$

if we can make the values of $f(x)$ arbitrarily close to L by taking x sufficiently very large. In other words, for every $\varepsilon > 0$, there is a number $N > 0$, such that

$$|f(x) - L| < \varepsilon \text{ whenever } x > N.$$

Similarly, we say that

$$\lim_{x \rightarrow -\infty} f(x) = L$$

if we can make the values of f arbitrarily close to L by taking x sufficiently very small. In other words, for every $\varepsilon > 0$, there is a number $N < 0$, such that

$$|f(x) - L| < \varepsilon \text{ whenever } x < N.$$

Horizontal Asymptote

The line $y = L$ is a horizontal asymptote of the graph of a function f if

$$\lim_{x \rightarrow \infty} f(x) = L \quad \text{or} \quad \lim_{x \rightarrow -\infty} f(x) = L$$

Theorem 11

(a) If r is a positive rational number and a is a real number, then $\lim_{x \rightarrow \infty} \frac{a}{x^r} = 0$.

(b) If r is a positive rational number, a is a real number, and x^r is defined for all x then $\lim_{x \rightarrow -\infty} \frac{a}{x^r} = 0$.

Example: It follows from the theorem that

$$\lim_{x \rightarrow \infty} \frac{3}{\sqrt[5]{x^4}} = \lim_{x \rightarrow \infty} \frac{3}{x^{4/5}} = 0.$$

Example: Graph $f(x) = \arctan(x)$, and list all horizontal asymptotes.

Infinite limit at infinity

A function $f(x)$ is said to have the infinite limit as x approaches infinity and we write

$$\lim_{x \rightarrow \infty} f(x) = \infty$$

if for every $M > 0$, there exists $N > 0$, such that,

$$f(x) > M \text{ whenever } x > N.$$

Remark We see that intuitively,

$$\lim_{x \rightarrow \infty} f(x) = \infty$$

if as we move along the x -axis, toward ∞ , value of $f(x)$ moves toward ∞ .

In general, if $r > 0$ is a rational number, then

$$\lim_{x \rightarrow \infty} x^r = \infty.$$

For instance,

$$\lim_{x \rightarrow \infty} \sqrt[3]{x^2} = \lim_{x \rightarrow \infty} x^{2/3} = \infty.$$

Theorem 12 Let $f(x) = a^x$, where $a > 0$ and $a \neq 1$, then

$$\lim_{x \rightarrow -\infty} a^x = \begin{cases} 0 & \text{if } a > 1 \\ \infty & \text{if } 0 < a < 1 \end{cases}$$

and

$$\lim_{x \rightarrow \infty} a^x = \begin{cases} 0 & \text{if } 0 < a < 1 \\ \infty & \text{if } a > 1 \end{cases}$$

In particular,

$$\lim_{x \rightarrow -\infty} e^x = 0 \text{ and } \lim_{x \rightarrow \infty} e^x = \infty.$$

Example 39 Find the following limits:

$$(a) \lim_{x \rightarrow \infty} (x^2 - x) =$$

$$(b) \lim_{x \rightarrow \infty} \left(6 - \frac{7}{x^3}\right) =$$

$$(c) \lim_{x \rightarrow -\infty} (5 - 3e^x - 2 \arctan x) =$$

Theorem 13 $\lim_{x \rightarrow \infty} \frac{\sin x}{x} = 0$, $\lim_{x \rightarrow -\infty} \frac{\sin x}{x} = 0$, $\lim_{x \rightarrow \infty} \frac{\cos x}{x} = 0$, and $\lim_{x \rightarrow -\infty} \frac{\cos x}{x} = 0$.

Example 40 Find the following limits:

$$(a) \lim_{x \rightarrow \infty} \frac{x + 3}{4x - 1}$$

$$(b) \lim_{x \rightarrow \infty} \frac{2x^2 + 3x + 5}{4x^2 - 2x + 1}$$

$$(c) \lim_{x \rightarrow \infty} \frac{x^2 + x - 1}{x + 2}$$

We generalize the above observations as follows:

Theorem 14 Let $f(x) = \frac{p(x)}{q(x)}$, where $p(x)$ and $q(x)$ are polynomials.

(a) If $\text{degree}(p) = \text{degree}(q)$ then

$$\lim_{x \rightarrow \pm\infty} \frac{p(x)}{q(x)} = \frac{\text{lcf}[p(x)]}{\text{lcf}[q(x)]}$$

(b) If $\text{degree}(p) < \text{degree}(q)$ then

$$\lim_{x \rightarrow \pm\infty} \frac{p(x)}{q(x)} = 0.$$

(c) If $\text{degree}(p) > \text{degree}(q)$ then

$$\lim_{x \rightarrow \pm\infty} \frac{p(x)}{q(x)} = \infty \text{ or } -\infty \text{ depending on the signs of lcf } p(x) \text{ and lcf } q(x).$$

Example 41 Find all horizontal asymptotes to

(a) $f(x) = \sqrt{4x^2 + 1} - 2x$

(b) $f(x) = \frac{x^2 + 2x \sin x + 3x + 4}{3x^2 - x \cos x - 9}$

$$(c) \ f(x) = \frac{2x^2 + \sqrt{x^4 - 2x^2}}{2x^2 + \sqrt{9x^4 - 2x^2}}$$

$$(d) \ f(x) = \frac{2x + \sqrt{x^2 - 2}}{2x + \sqrt{9x^2 - 2}}$$

Example 42 Find the following limits

$$(a) \lim_{x \rightarrow \infty} \frac{2x^3 - 4x^2 + 9}{4x^3 - 1} =$$

$$(b) \lim_{x \rightarrow \infty} \frac{2 - e^x}{1 + 2e^x} =$$

$$(c) \lim_{x \rightarrow \infty} \frac{2e^{3x} - e^{-5x}}{1 + 4e^{3x}} =$$

$$(d) \lim_{x \rightarrow -\infty} \frac{2e^{-5x} - e^{3x}}{e^{4x} + 2e^{-5x}} =$$

$$(e) \lim_{x \rightarrow -\infty} \frac{-2x + 3}{\sqrt{x^2 + 4}} =$$

$$(f) \lim_{x \rightarrow \infty} \arctan \left(\frac{\sqrt{2x^2 + 1}}{2x - 1} \right) =$$

$$(g) \lim_{x \rightarrow \infty} \ln \left(\frac{2x^2 + 5}{x^2 + 4} \right) =$$

$$(h) \lim_{x \rightarrow -\infty} \left(\sqrt{x^2 + 9} - x \right) =$$

$$(i) \lim_{x \rightarrow \infty} [\ln(3x + 1) - \ln(3 + x)] =$$

Lecture 11: Derivative

Derivative of a Function at a Point: Let f be defined on an open interval that contains a . The derivative of a function f at $x = a$ is denoted by $f'(a)$, and is defined as:

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \quad (1)$$

provided the limit exists.

If we write, $x = a + h$, so that $x - a = h$ and $h \rightarrow 0$ implies $x \rightarrow a$, then $f'(a)$ becomes

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}.$$

This is an alternate form of the derivative of $f(x)$ at $x = a$.

Example 43 For given function $f(x)$, find the indicated derivatives:

(a) $f(x) = x^2$, find $f'(3)$.

(b) $f(x) = \sqrt{x}$, find $f'(4)$

(c) $f(x) = \frac{2}{x-1}$, find $f'(5)$

Meaning of Derivative:

1. $f'(a)$ = the slope of tangent line to the graph of $y = f(x)$ at the point $(a, f(a))$.
2. $f'(a)$ = the instantaneous velocity of a moving object at time $t = a$ if $f(t)$ represents its position.

Example 44 Find an equation of tangent line to the graph of $y = \frac{1}{2}x^2$ at the point $P(2, 2)$

Example 45 A rock falls from a 890 ft bridge above the level of the water. The height (in feet) of the rock falls after t seconds is given by $f(t) = 890 - 16t^2$,

- (a) Find average velocity of the rock on the time interval $[1, 3]$.
- (b) Find the instantaneous velocity of the rock at time $t = 3$ seconds.